3) A health nutritionist wanted to compare the effects of three popular diets, together with regular physical exercise provided by personal trainers, on the weight loss of overweight individuals. The three popular diets are:

Diet 1: The Zone diet, where all daily meals and snacks are delivered to the individual.
Diet 2: Weight Watchers' Special diet
Diet 3: A low carb-high protein diet.

For the purpose of the study a random sample of 4 personal trainers were picked from a population of such trainers. (A personal trainer works on a one-to-one basis with an individual on an exercise program for two hours each week.) The primary goal of the study is to see if there is statistically significant evidence of a difference in the weight loss of the diets when averaged over the population of personal trainers.

A random sample of 60 overweight individuals were picked for the study, and assigned completely at random, 15 to each trainer. The 15 individuals for each trainer were assigned completely at random, 5 to each of the three diets. Thus there were 5 overweight individuals assigned to each of the 12 combinations of 4 trainers and 3 diets. After three months under the diet/trainer regimen the following **mean** weight losses (in pounds) were observed for each of the 12 diet/trainer combinations. Each mean was based on 5 observations.

	Trainer 1	Trainer 2	Trainer 3	Trainer 4
Diet 1	19	24	27	22
Diet 2	21	13	18	24
Diet 3	14	14	21	23

The average of the 12 cell sample means = 20.
The average of the 12 cell sample variances = 19.5

$$19^2+24^2+27^2 + 22^2 +21^2+13^2 +18^2 +24^2 + 14^2+14^2+21^2+23^2 = 5022$$

$$92^2 + 76^2 +72^2 = 19424 \quad ; \qquad 54^2 + 51^2 +66^2 + 69^2 = 14634$$

Complete the ANOVA table below. (Assume that the standard assumptions of normality, homogeneity of variance, and independence hold). Carry out all tests at the 0.05 level of significance, and show the critical Tabled F-values. State your conclusions.

Source	df	SS	MS	F-ratios
Diet				
Trainer				
Diet x Trainer				
Error				
Total				